MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION MARCH 2026

FOURTH SEMESTER

Skill Enhancement Course (SEC) - MG4SECBTG200 – QUALITY CONTROL IN BIOLOGY

(2024 ADMISSION ONWARDS)

Duration: 1.5 Hours **Maximum Marks: 50**

Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A One word answer

Answer any 10 questions. Each question carries 2 marks.

1. Name any 1 regulatory agency involved in quality control. [K] / [CO4]
2. Identify a software tool commonly used in quality control. [K] /[CO3]
3. State 1 application of using ATP bioluminescence in testing quality. [A] / [CO3]
4. Expand HACCP. [K] / [CO4]
5. Define quality assurance. [U] / [CO2]
6. Define Donabedian model of quality. [U] / [CO1]
7. Give an example of an immunological tool used in quality control. [K] / [CO3]
8. Give an example of a food safety certification. [K] / [CO4]
9. State a best practice for quality control in electrophoresis. [U] / [CO3]
10. Define quality control. [U] / [CO1]
11. Recall a commonly used sterilization method. [U] / [CO3]
12. State what you mean by CCP. [U] / [CO4]

(10 x 2 = 20)

Part B Short essay

Answer any 5 questions. Each question carries 4 marks.

13. Recall the benefits of quality control in PCR. [U] / [CO3]
14. Tabulate the differences between QA and QC. [U] / [CO2]
15. Outline the importance of documentation in quality control. [A] / [CO5]

16. Describe the good manufacturing practices to be followed in food industry. [U] / [CO3]
17. Recall what you mean by ICH guidelines. [U] / [CO5]
18. Discuss briefly about Food Safety Modernisation Act (FMSA). [U] / [CO4]
19. Discuss on why quality control is considered a multidisciplinary approach. [U] / [CO2]

(5 x 4 = 20)

Part C Essay

Answer any 1 question. Each question carries 10 marks.

20. Describe the 7 principles of HACCP. [U] / [CO4]
21. Describe the different methods of sterilization and its validation. [U] / [CO3]

(1 x 10 = 10)